IBM UNIVERSITY ENGAGEMENT PROJECT SUBMISSION

“Empowering Customer Support Analytics with IBM watsonx.ai,
IBM Granite Models, and IBM BOB”

Project Tile	**AI Customer Support Analytics using IBM watsonx.ai Granite & IBM BOB**
Domain of Project	**Customer Support / Enterprise AI**

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Problem Statement

Problem Statement : AI Customer Support Analytics

The Challenge

Organizations receive thousands of **customer support requests** every day through **emails, chats, and support tickets**. Manually analyzing these conversations is **time-consuming** and often fails to **identify customer sentiment, recurring issues, and business risks**. As a result, companies struggle to improve customer satisfaction and agent performance. An AI-powered solution is required to **automate customer support analytics and generate actionable business insights**.

The Objective

The objective of this project is to develop an AI-powered customer support analytics platform using **IBM watsonx.ai Granite Models** and **IBM BOB**. The system automatically analyzes customer conversations, identifies customer sentiment, detects emotions and intents, evaluates support agent performance, predicts churn risk, and generates business recommendations. This enables organizations to make faster and more informed decisions while improving customer experience:

- Automate customer support analytics using IBM Granite Models.
- Generate actionable business insights.
- Improve customer satisfaction and agent productivity.

Proposed Solution

Proposed Solution: AI Customer Support Analytics

The proposed solution is a Flask-based web application that uses **IBM watsonx.ai Granite Models** to analyze customer support conversations. Users can upload customer support datasets in **CSV or Excel** format, and the system performs **AI-driven sentiment analysis, emotion detection, intent recognition, issue categorization, satisfaction scoring, and churn prediction**. The results are displayed through an **interactive dashboard with charts, KPIs, and downloadable PDF and Excel reports**, helping businesses quickly understand customer concerns and operational performance.

The platform performs several AI-driven tasks:

- **Sentiment Analysis Module:** identifies whether customer conversations are Positive, Neutral, Negative, or Very Negative, helping organizations understand customer satisfaction trends.
- **Emotion & Intent Detection Module:** recognizes customer emotions such as Angry, Frustrated, Happy, or Confused while also identifying the customer's intent, including refund requests, billing issues, technical problems, or product inquiries.
- **Issue Classification & Risk Prediction Module:** automatically categorizes customer issues, predicts ticket priority and urgency, estimates customer satisfaction scores, detects churn risk, and identifies conversations that require escalation.
- **Agent Performance Evaluation Module:** analyzes support agent responses by measuring professionalism, empathy, clarity, grammar, friendliness, and resolution quality. It also provides AI-generated suggestions to improve future customer interactions.

Technology Used

IBM watsonx.ai: Provides access to IBM Granite foundation models for AI-powered analysis. It is used to perform sentiment analysis, emotion detection, intent recognition, issue categorization, and business insight generation.

IBM Granite Model (e.g. meta-llama/llama-3-3-70b-instruct): The core AI model used to understand customer conversations, reason over support tickets, generate executive summaries, evaluate agent performance, and produce actionable business recommendations.

IBM Cloud: Provides a secure and scalable cloud environment for deploying the application and integrating IBM watsonx.ai services. It also manages authentication and API access for Granite models.

IBM BOB: Used during project development to accelerate coding, debugging, application design, and deployment. IBM BOB helped improve development productivity and streamline the implementation process.

Python: The primary programming language used for backend development, AI integration, data processing, and business logic implementation.

Flask: A lightweight Python web framework used to build the web application, manage routing, handle file uploads, and serve the analytics dashboard.

Pandas: Used for reading, cleaning, validating, and preprocessing customer support datasets before AI analysis.

Chart.js: Creates interactive charts and visualizations such as sentiment distribution, issue categories, customer satisfaction, and agent performance dashboards.

Bootstrap 5, HTML, CSS & JavaScript: Used to build a responsive, modern, and user-friendly interface for displaying analytics and reports.

ReportLab & OpenPyXL: Generate professional PDF reports and Excel workbooks containing AI-generated insights and detailed analytics.

AI Workflow Components Used-

1) File Upload Component - Allows users to upload customer support datasets in **CSV, XLS, or XLSX** format. The component validates the uploaded file and prepares it for AI-powered analysis.

2) Data Processing Component - Preprocesses and cleans the uploaded data using **Python** and **Pandas**. It performs data validation, column normalization, missing value handling, and formatting to ensure accurate AI analysis.

3) IBM watsonx.ai Component - Connects the application to **IBM watsonx.ai**, enabling secure access to IBM Granite foundation models for intelligent customer support analytics and business insight generation.

4) IBM Granite Model (meta-llama/llama-3-3-70b-instruct) - Analyzes customer support conversations to perform **sentiment analysis, emotion detection, intent recognition, issue categorization, priority prediction, customer satisfaction scoring, churn risk assessment, and executive summary generation**.

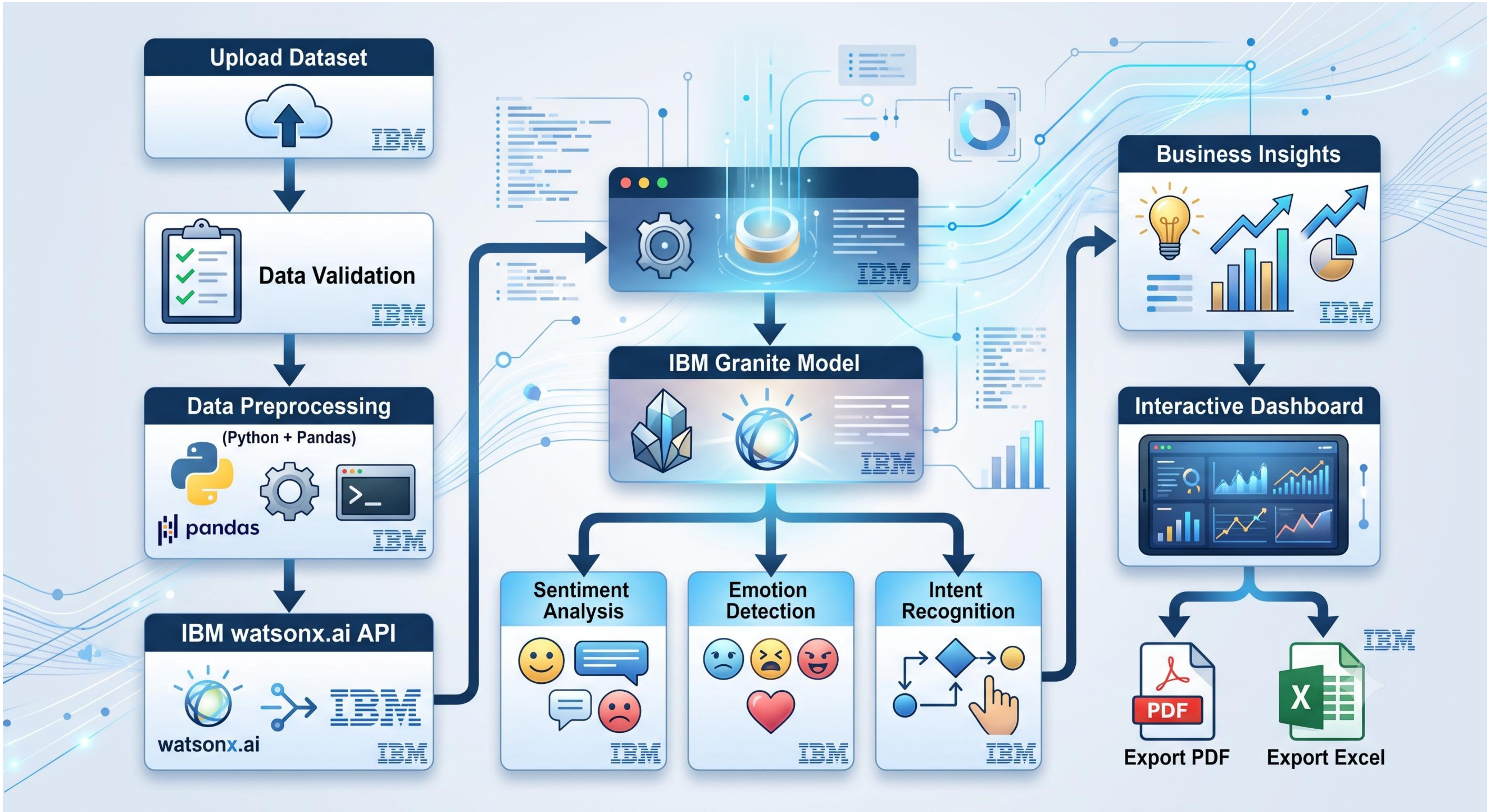
5) Analytics Dashboard Component - Displays AI-generated insights through an interactive dashboard featuring **KPI cards, sentiment distribution, issue category charts, agent performance metrics, customer satisfaction trends, business risks, and executive recommendations**.

6) Export & Reporting Component - Generates professional reports in **PDF, Excel, CSV, and JSON** formats, allowing organizations to download, share, and analyze customer support insights efficiently.

Application Workflow

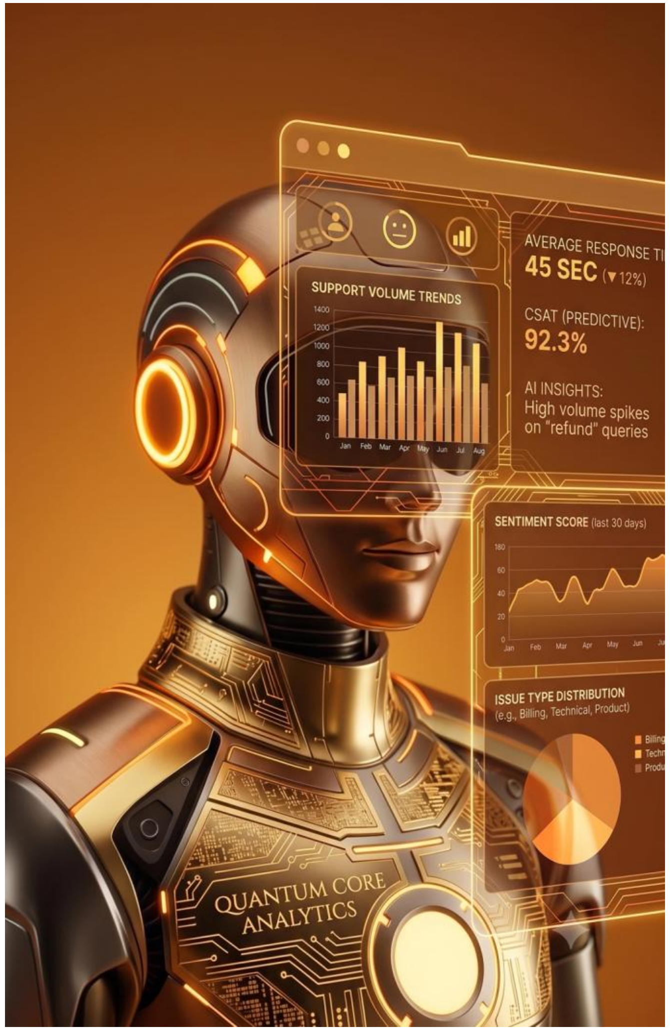
Customer Support Dataset → File Upload → Data Processing → IBM watsonx.ai → IBM Granite Model → AI Analysis → Interactive Dashboard → Export Reports

Architecture Workflow-



Role of Artificial Intelligence in the Solution

Role of Artificial Intelligence in AI Customer Support Analytics



Artificial Intelligence is the core technology that powers the Customer Support Analytics platform. Using **IBM watsonx.ai** and **IBM Granite Models**, the system automatically analyzes customer support conversations and converts unstructured text into meaningful business insights.

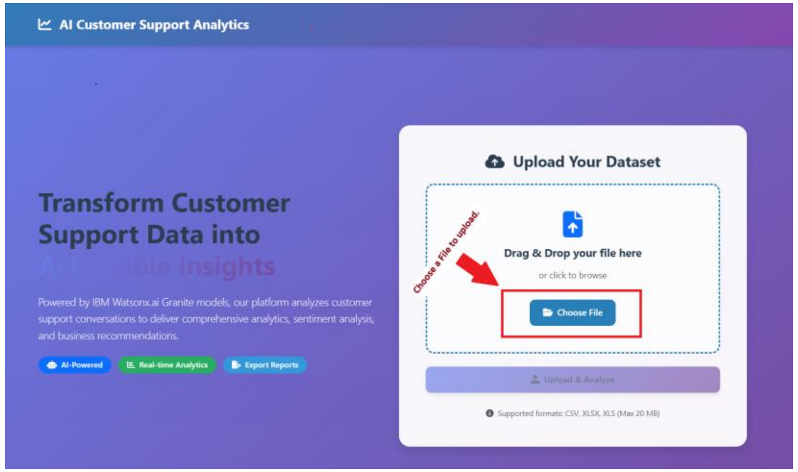
The AI identifies **customer sentiment**, **emotions**, **intent**, **issue categories**, **ticket priority**, **urgency**, **customer satisfaction scores**, and **churn risk**. This enables organizations to quickly understand customer concerns, identify recurring issues, and prioritize critical support requests without manual effort.

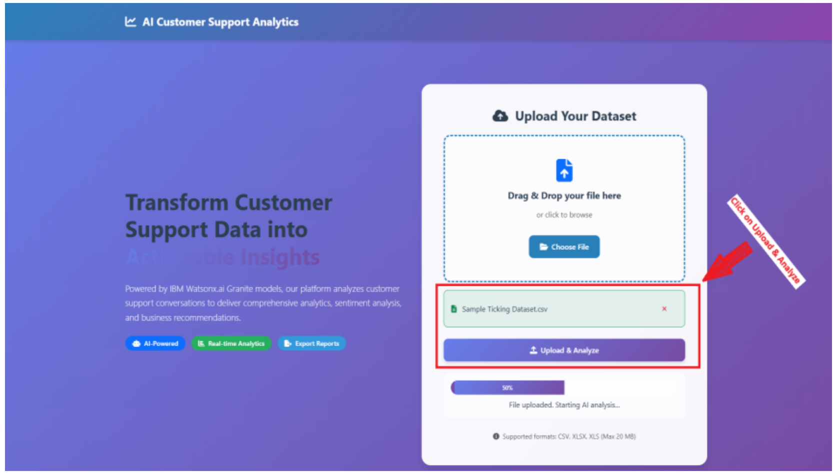
In addition to customer analysis, the AI evaluates **support agent performance** by assessing professionalism, empathy, communication clarity, grammar, friendliness, and resolution quality. It also generates executive summaries, identifies operational bottlenecks, highlights business risks, and provides actionable recommendations to improve customer service operations.

The processed insights are presented through an interactive analytics dashboard with KPI cards, visual charts, and downloadable PDF and Excel reports, enabling managers to make faster, data-driven decisions.

Overall, Artificial Intelligence transforms customer support data into valuable business intelligence, helping organizations improve customer satisfaction, enhance agent performance, reduce operational costs, and optimize decision-making.

How it Works

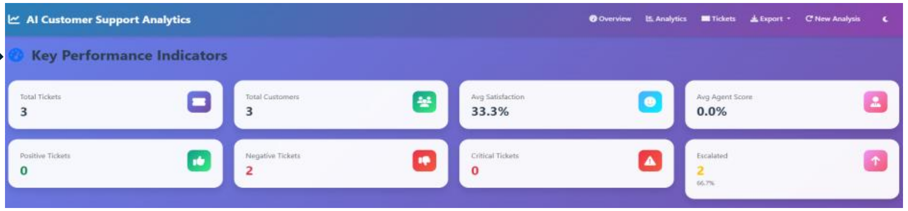


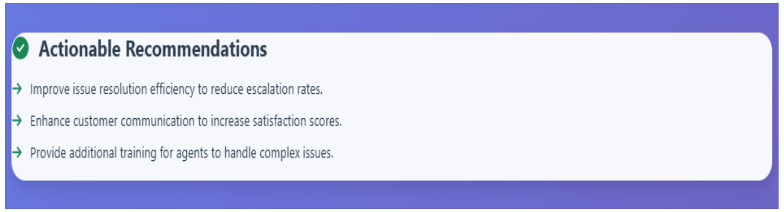


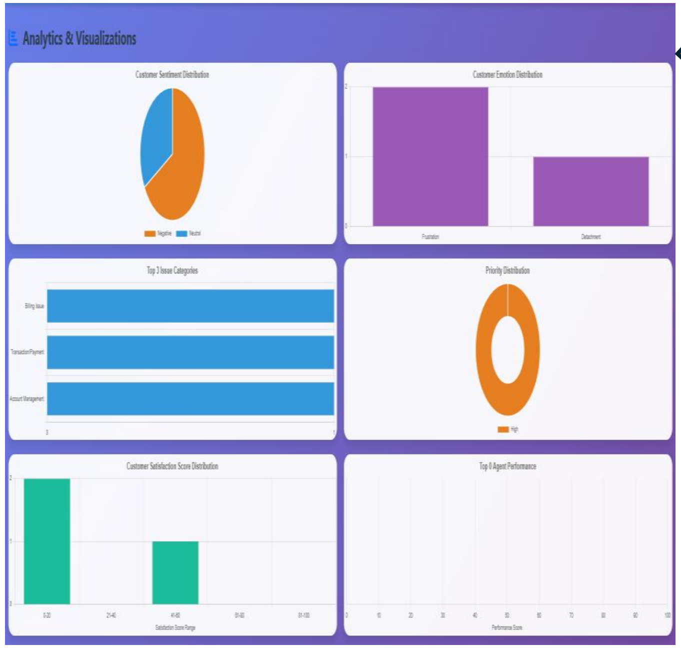
Choose a Dataset file to upload

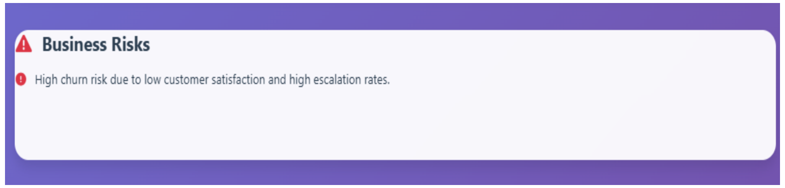
Click on Upload & Analyze

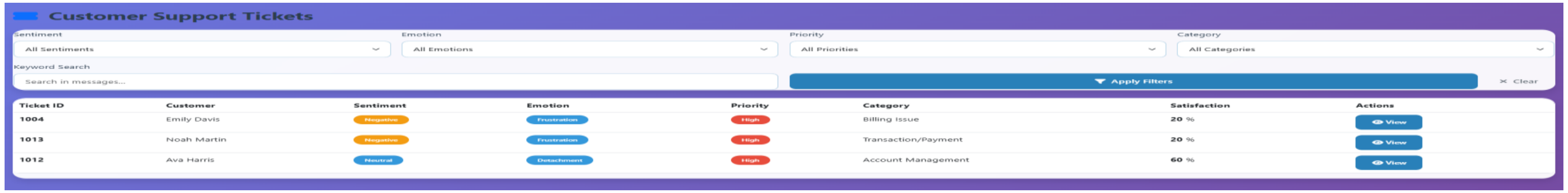
This will automatically generate KPIs, Full Dashboards, Summary, Recommendations, Business Risks, Customer Support Tickets.







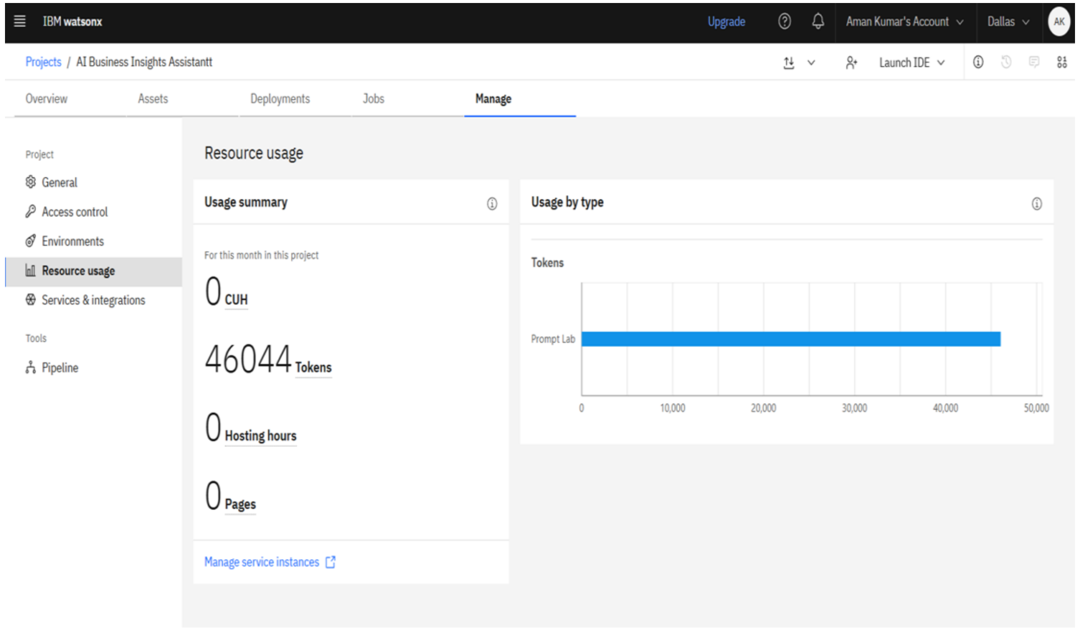




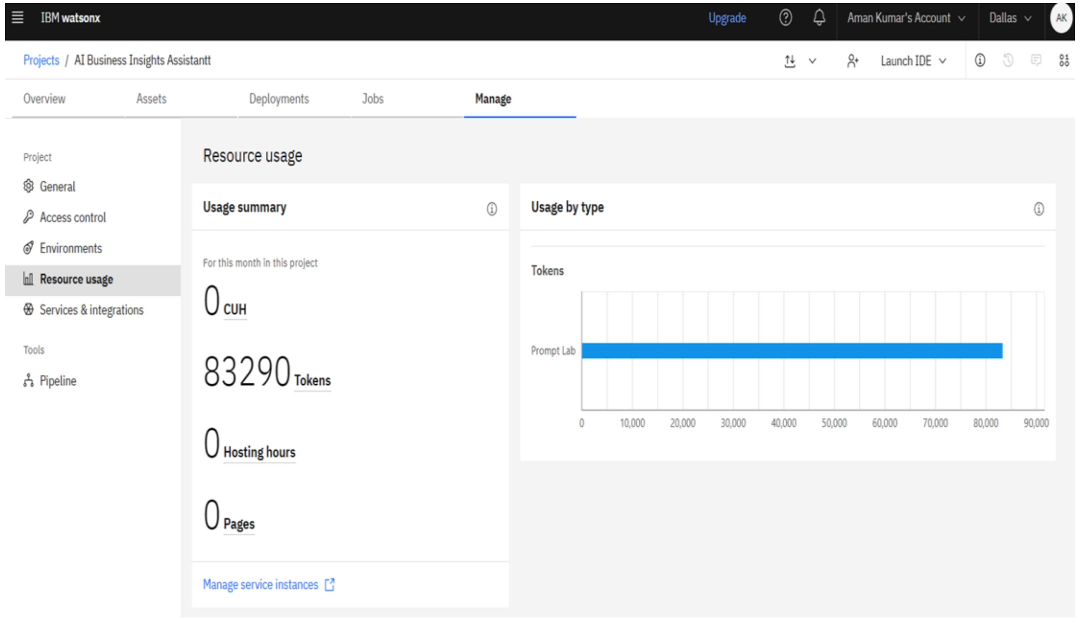
Token Usage

37,246 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed **AI Customer Support Analytics** platform stands out by combining **Artificial Intelligence, Business Intelligence, and Customer Support Analytics** into a single intelligent solution powered by **IBM watsonx.ai**, **IBM Granite Models**, and **IBM BOB.**

- **AI-Powered Customer Conversation Analysis:** Instead of manually reviewing customer support tickets, the platform automatically performs **sentiment analysis, emotion detection, intent recognition, issue categorization, and priority prediction** using IBM Granite Models.
- **Comprehensive Customer Support Intelligence:** The solution provides end-to-end analytics by combining customer sentiment, satisfaction scoring, churn risk assessment, escalation detection, and business insights within a single platform.
- **AI-Based Agent Performance Evaluation:** Unlike conventional dashboards, the platform evaluates support agents based on **professionalism, empathy, clarity, grammar, friendliness, & resolution quality**, while also providing AI-generated improvement suggestions.
- **Interactive Business Intelligence Dashboard:** The application transforms complex customer support data into intuitive KPI cards, charts, executive summaries, operational bottlenecks, business risks, and actionable recommendations for faster decision-making.
- **Automated Report Generation:** The platform generates professional **PDF, Excel, CSV, and JSON** reports automatically, reducing manual reporting effort and enabling easy sharing of business insights.
- **Enterprise-Ready and Scalable:** Built using **IBM watsonx.ai**, **IBM Granite Models**, **IBM Cloud**, **Flask**, and **Python**, the solution is secure, scalable, and can be integrated into enterprise customer support environments.
- **Real-World Business Impact:** The platform enables organizations to reduce manual analysis, improve customer satisfaction, optimize agent performance, identify recurring issues, and make faster, data-driven business decisions.

GitHub Repository

Repository Name	GitHub Repository URL
AI-Customer-Support-Analytics	https://github.com/AmaanOO7/AI-Customer-Support-Analytics

Deployed Project Name	Deployed Link
AI-Customer-Support-Analytics	https://ai-customer-support-analytics.onrender.com/

Repository Contents	Technologies Used	Project Highlights
✓ README.md	IBM watsonx.ai	AI-powered Customer Support Analytics
✓AGENT_INSTRUCTIONS.md	IBM Granite Models	Sentiment & Emotion Detection
✓ app.py	IBM Cloud	Intent Recognition
✓ requirements.txt	IBM BOB	Customer Satisfaction Analysis
✓ .env.example	Python	Agent Performance Evaluation
✓ IBM Project (.pptx)	Flask	Interactive Dashboard
✓ IBM Project (.pdf)	Pandas	Export Reports (PDF, Excel, CSV, JSON)
✓ Problem Statement (.pdf)	Chart.js	
✓ Sample Dataset	Bootstrap	**Project Video**
✓ Project Screenshots	Report Lab	**https://drive.google.com/file/d/1SbouEY3sLHENnbWPDsQXn7OC36XTuHjy/view?usp=sharing**
✓ .gitignore		
✓ .python-version		
✓ Project source files		

Future Scope

1. **Real-Time Customer Support Analytics -** The platform can be integrated with live customer support systems such as **Zendesk, Freshdesk, Salesforce Service Cloud, and Microsoft Dynamics 365**. This will enable real-time analysis of customer conversations, allowing organizations to detect critical issues instantly, prioritize support requests, and improve response times.

2. **Multilingual & Voice-Based Customer Support -** Future versions can support **multiple languages** and **voice conversation analysis**. By integrating speech-to-text technology and multilingual IBM Granite Models, the platform will be able to analyze customer calls, chat transcripts, and emails from global customers, making the solution more accessible for international organizations.

3. **Predictive Analytics & Smart Ticket Routing -** Machine learning models can be incorporated to predict customer churn, estimate ticket resolution time, and automatically route support tickets to the most suitable agent based on expertise, workload, and issue category. This will improve operational efficiency and reduce resolution time.

4. **CRM & Enterprise Integration -** The solution can be integrated with enterprise CRM platforms, helpdesk software, and communication tools such as Slack and Microsoft Teams. This will allow organizations to receive AI-generated insights directly within their existing workflows without switching applications.

5. **Advanced AI & Generative Assistance -** Future enhancements can include AI-powered response suggestions, automatic ticket summarization, knowledge base recommendations, and intelligent reply generation using advanced IBM Granite Models. These capabilities will further improve agent productivity and customer satisfaction.

6. **Enterprise-Scale Deployment -** The application can be deployed on IBM Cloud with support for secure authentication, user management, role-based access control, and large-scale analytics. This will enable organizations to analyze millions of customer interactions while maintaining performance, security, and scalability.

Certificates Earned:

In recognition of the commitment to achieve professional excellence

Getting Started with Artificial Intelligence
IBM SkillsBuild

Aman Kumar

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Jun 24, 2026
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/2b1fd9a2-1627-430a-af17-a7e0bab46b55

IBM

Certificates Earned:

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Aman Kumar

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 29 Jun 2026 (GMT)

Learning hours: 30 mins

Certificates Earned during Internship Period:

Certificated Earned	Link
Create a Credly account	
Cybersecurity with Gen AI	
Exploring Quantum Computing cert	
IBM Artificial Intelligence Certificate	
IBM Cybersecurity Certificate	
Troubleshoot Your Code Using IBM Bob	
Edunet AICTE Intership	

Thank You!